

CONDUCTOMETRIC ESTIMATION OF MIXTURE OF STRONG AND WEAK ACID (CONDUCTOMETRIC SENSORS)

Aim: Estimation of strong acid and weak acid from given mixture by conductometric titration.

Principle:- Reciprocal of resistance (R) is conductance (L) and is measured in Siemens.

$$L = (1/R)$$

A conductivity cell consists of a pair of platinized foils of definite distance ' l ' apart. Thus ' l/a ' of given conductivity cell has a definite value and it is known as cell constant.

Specific Conductance^{Electrolyte} is the conductance of an ~~the~~ solution, placed between two parallel electrodes 1cm apart and area of cross section 1cm^2 .

Equivalence Conductance is the conductance of the solution due to all the ions produced by dissolving one-gram equivalent weight of a solute in 1000cm^3 of a solution.

Molar Conductance is the conductance of the solution due to all the ions and number of ions produced by dissolving one-gram molecular weight of a solute in 1000cm^3 of a solution.

Conductance of an electrolyte is directly proportional to the mobility of ions and number of ions present in unit volume. Conductance measurements can be employed to find the

End points of acid-base in the burette.

During titration, initially highly mobile H^+ ions in the solution are replaced by less mobile Na^+ ions. Therefore, there is a decrease in conductance of the solution as more and more $NaOH$ solution is added to neutralize HCl .

This continues until HCl gets neutralized by the base. After the endpoint, further addition of $NaOH$ sol. result in neutralization of the acetic acid and conductance increase marginally due to that formation of salt, sodium acetate, which dissociates marginally better than acetic acid. After the neutralization of acetic acid, conductance increases significantly due to unused sodium hydroxide that furnishes Na^+ and highly mobile OH^- ions.

The conductance of the solution is plotted against the volume of the base added. On extrapolation of the three straight lines give two intersections. The volume of $NaOH$ corresponding to first intersection gives neutralization point of CH_3COOH .